5.02	Direct and inverse proportion				
5.02a	Direct proportion	Solve simple problems involving quantities in direct proportion including algebraic proportions. e.g. Using equality of ratios, if $y \propto x$, then $\frac{y_1}{y_2} = \frac{x_1}{x_2}$ or $\frac{y_1}{x_1} = \frac{y_2}{x_2}$. Currency conversion problems. *[see also Similar shapes, 9.04c]*	Solve more formal problems involving quantities in direct proportion (i.e. where $y \propto x$). Recognise that if $y = kx$, where k is a constant, then y is proportional to x.	Formulate equations and solve problems involving a quantity in direct proportion to a power or root of another quantity.	R7, R10, R13
5.02b	Inverse proportion	Solve simple word problems involving quantities in inverse proportion or simple algebraic proportions. e.g. speed–time contexts (if speed is doubled, time is halved).	Solve more formal problems involving quantities in inverse proportion (i.e. where $y \propto \frac{1}{x}$). Recognise that if $y = \frac{k}{x}$, where k is a constant, then y is inversely proportional to x.	Formulate equations and solve problems involving a quantity in inverse proportion to a power or root of another quantity.	R10, R13

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to…	Foundation tier learners should also be able to…	Higher tier learners should additionally be able to…